

DAVID XU

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Education

University of Toronto: Bachelor of Applied Science in Civil Engineering + PEY

Sep 2024 - Apr 2029

Certifications

Software

- | | | | | | |
|----------------------|--|-------------------|----------|--------------|-------------------|
| • SFA & CPR C (2024) | • Rhinoceros 3D | • Microsoft Suite | • ArcGis | • QGIS | • MicroStation |
| • WHMIS (2026) | • Autodesk Suite (Autocad, Revit, etc) | • Adobe suite | • Vscope | • Solidworks | • Ansys Workbench |

Work Experience

Trenton Cadet Training Centre (Cadet Correspondent)

Jun 2025 - Aug 2025

- Worked with the Brigadier General to align the Cadet Program with her strategic vision for the program development
- Managed internal and external information flow, ensuring operational security and a professional public representation
- Filled in role of officer
- Workflow ~3000 photos per day using excel for audit archive and the quota of 3 posts and 1 video project a day for 10 weeks

Decathlon Canada (Store Technician)

Feb 2023 - Jul 2024

- Provided service, maintenance, training for all sports, store equipment, and to all 5 store section teams
- Built an efficient customer response and delivery workflow that increased service time by 10-fold in the workshop
- Encouraged store tradition amongst colleagues to improve store moral and to increase service levels and profit margins

Lewis Design and Construction (Internship Student)

Jul 2023 - Aug 2023

- Created and review floor plans and performed site analyses to ensure design accuracy, consistency and meeting client needs
- Utilized Blender to create high-quality renderings for presentation, delivering significant cost and time savings
- Collaborated with senior to refine concepts and ensure project feasibility and compliance with regulations

Experience and Projects

NEIGES Biotechnologies Well Plate Video Module (CAD Specialist)

Feb 2026 - Present

- Assisting the development of CellSnooper, an in-incubator device for real-time cell contractility monitoring
- Designed modular, 3D-printed structures using off-the-shelf components with multiple degrees of freedom for calibration and imaging
- Collaborated remotely with a cross-university team on the second physical prototype, designing with predefined electronic components

Grand River Watershed Delineation - ArcGIS Pro (Academic Project)

Jan 2026- Feb 2026

- Delineated the 6,782 km² Grand River watershed using hydrologic tools (Fill, Flow Direction, Flow Accumulation, Watershed)
- Processed and projected DEM data (WGS 1984 UTM Zone 17N) for accurate elevation and drainage analysis
- Analyzed watershed characteristics including elevation, perimeter, and drainage network topology
- Generated drainage boundaries and stream networks; validated outputs against basemap/hydrography layers
- Produced map layouts with consistent symbology, labeling, legend, scale bar, and export settings
- Managed feature classes and attribute tables; maintained organized geodatabase structure and naming conventions

Building Takedown Estimation - Revit (Academic Project)

Oct 2025 - Nov 2025

- Developed a BIM-based quantity takeoff and cost estimate for a residential building using Revit and RSMeans covering 200+ materials
- Built a structured method to select between RSMeans Unit and Assembly pricing methods, avoiding double counting
- Produced an industry-style Excel executive summary consolidating trade costs, percentages, and cost per square metre for review

Queen's Park Station Capacity Report (Academic Project)

Oct 2025 - Nov 2025

- Led congestion assessment and decision analysis for the Queen's Park Station Northwest exit bottleneck
- Modeled 9×1.8 m (16.2 m²) corridor with ≤6 ppl/m² comfort; compared Pathway Expansion, North Exit, and FC Tunnel
- Recommended Pathway Expansion (~5.0 m) + North Exit; meets ~250 peak at comfort, reducing density and egress time

FROSH Week SUBCOM (Videography Chair)

Mar 2025 - Feb 2026

- Managed and organized large volumes of content for event documentation and archival use for the largest FROSH Event in Canada
- Coordinated filming schedules and deliverables with organizing teams under weekly deadlines
- Maintained structured folder systems and file naming conventions to ensure accuracy and accessibility

UofT Autodrive Team (Video Specialist)

Oct 2025 - Present

- Collaborating with technical team to filmed and edited driving footage for post-drive analysis, documentation, and performance review
- Created video content for social media, outreach, and public-facing communications for recruiting future team members and sponsors
- Supported on-site testing operations, including drive course setup and general team assistance on top of managing a 5-camera setup

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Sub Tooth Blood Flow Detection (Multi-Modality Coordinator)

Jan 2025 - Present

- Working on new methods of blood flow detection within the pulp chamber with the Dental Faculty at UofT
- Reaching out to different departments and faculties to invent new method of subsurface flow detection
- Currently focused on a Lidar Flowmetry Technique pending further research on spectral stability

Field Survey Assistance (Self Interest)

Sep 2024 - July 2025

- Applied photogrammetry and gaussian splatting techniques for precise measurements and mapping through drone footage
- Utilized modeling programs such as Autodesk Recap and Blender to create models and prototypes for project visualization
- Following techniques were applied to help different disciplines to assist on campus projects at UofT

UofT Unmanned Aerial Systems SAE Team (Design Team Structural Subsystem)

Sep 2025 - Present

- Designed and assembled SolidWorks mechanism for payload handling with a team of 8 from across different disciplines
- Prototyped 5 single servo designs testing its structural capabilities, weight, ease of use, DFM/A, and ease of installation
- Designed fuselage in SolidWorks and prepared AutoCAD files for laser-cut with space optimization
- Achieved 4th place overall at the SAE competition, delivering a 3x increase in total points compared to the previous year

Royal Canadian Air Cadets (Squadron Warrant Officer)

Sep 2019 - Sep 2025

- Mentoring over 20 cadet leaders to develop over 150 cadets to become thoughtful teammates to excel together
- Lead of the PA team that outputs weekly videos and posts to gather engagement of the local community
- Boost in effective team interaction and moral resulting in one of the most well-known Corps in Canada

Bahen Courtyard Project (Academic Project - Project Manager)

Sep 2024 - Dec 2024

- Collaboration with the Sustainability office at UofT and group of 6 to transform a decommissioned premise in Bahen Centre of IT
- Full solution was implemented including floor plan and 3D Visualizations implemented in Blender for fast preliminary plan
- Transformed a decommissioned premise to a hustling “third place” and community gathering place

Bike Fork Mount (Personal project)

Dec 2023 - Jun 2024

- Designed robust frame for mounting on bike forks that aren't designed for mounting front pannier racks in Fusion 360 and tested on CAE
- Designed for manufacturing through CNC milling, Laser Cutting, or 3D printing allowing a fast, strong, and accessible product
- Adaptive front rack can increase bike payload by an extra 20 kilograms allowing flexible bike configurations for bike packing

YYZ-MTL Bike Packing Ride (Personal project)

8 days in Aug 2023

- Organized an 8-day ride with a group of 3 that spanned over 1200 Kilometer from across Ontario to Quebec
- Explored rural communities and military bases and interacted with its citizens and local infrastructure
- Helped gain an understanding of the current infrastructure status in this country

Hack the Ridge High School Hackathon (Outreach Coordinator)

Jul 2023 – Aug 2024

- Several educational workshops were run to guide newbies to obtain tools to compete effectively in the hackathon
- My promotion strategies provided a reach of over 400 sign-ups which was 3x the previous year's attendees
- Raised over \$5000 in Sponsorship money just for prizes, facilities, food, merchandise, volunteers

2022-2023 Yearbook (Secondary Editor)

Sep 2022 - Dec 2023

- Responsible for organizing a team over 30 people with over 120 pages of content within 4 months to create a consistent yearbook
- Organized design and content gathering workflow through a yearbook ladder for easy viewing and collaboration
- First to introduce uniform colour grading creating a more pleasant and organized viewing experience

Airbus A350F Livery Design Contest (Self Interest)

Sep 2022 – Nov 2022

- Drafted a creative conceptualization that was integrated innovative aesthetics and functional elements
- Contributed to a design that aligns with the mission for creativity and relevance in the aviation sector
- Ensured the livery design harmonized with Airbus' corporate identity, emphasizing branding

London Luton Airport Revitalization (Personal project)

Dec 2021 - Aug 2023

- Designed a long-term masterplan concept for an over-capacity airport located in the northern outskirts of London
- Created renderings for terminal upgrades and multimodal transport links to support new capacity requirements
- Explored highway and transit integration strategies with current transit options such as the DART and other connections
- Developed optimized rendering and design workflows to reduce computational load and increase productivity

Personal Interests: Skiing, Bikepacking, Photography/Videography, CNC Machines, Bird & Plane Spotting, Designing Gadgets